

Task 5: Session 5 (AIML) - Assignment Questions

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Domain: AIML

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1. String Indexing

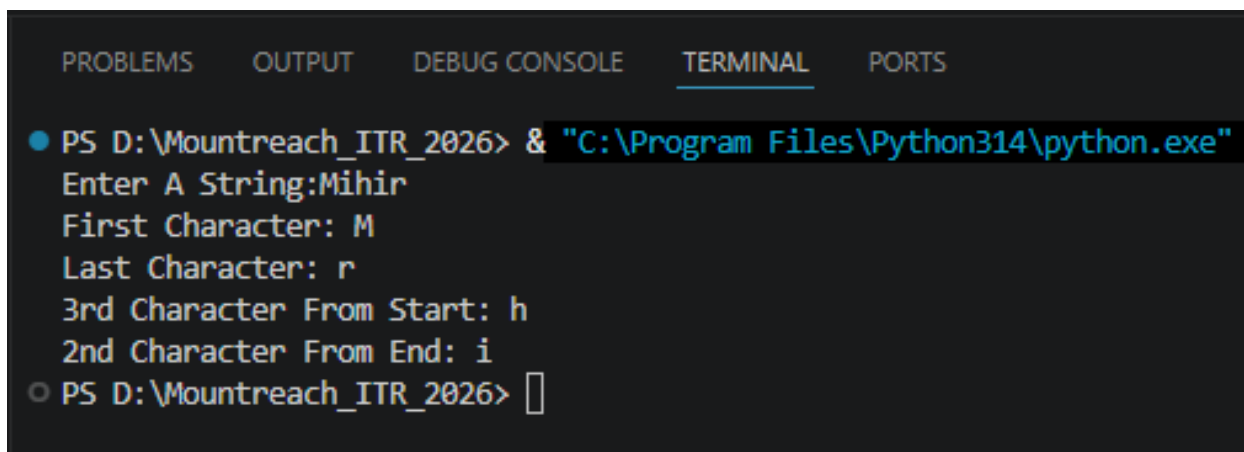
Write a program that takes a string input from the user.

Print:

- The first character
- The last character (using negative index)
- The 3rd character from the start
- The 2nd character from the end

Add comments explaining positive and negative indexing.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:Mihir
First Character: M
Last Character: r
3rd Character From Start: h
2nd Character From End: i
○ PS D:\Mountreach_ITR_2026> 
```

2. String Slicing

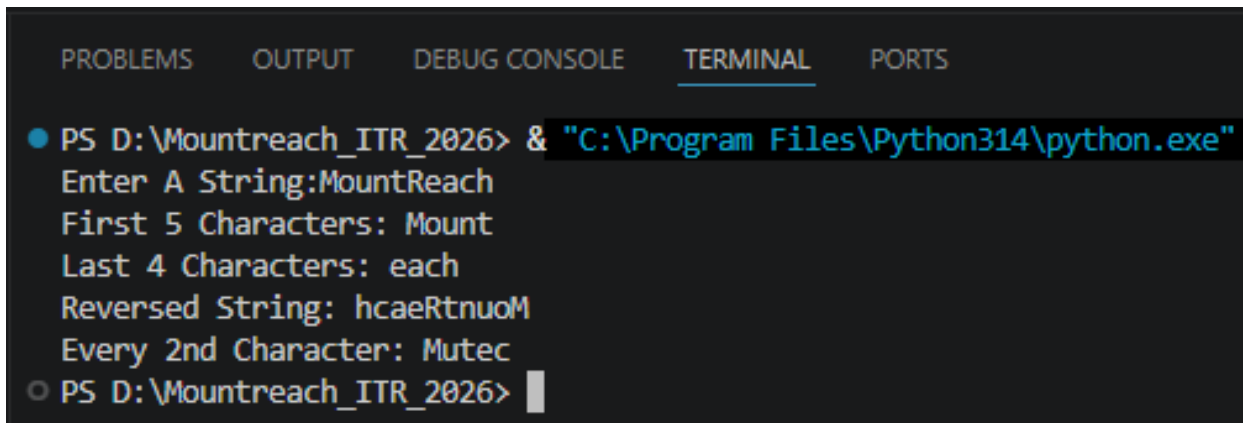
Take a string input from the user.

Using slicing, print:

- The first 5 characters
- The last 4 characters
- The string in reverse order
- Every 2nd character of the string

Include comments explaining the slicing syntax [start:end:step].

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:MountReach
First 5 Characters: Mount
Last 4 Characters: each
Reversed String: hcaeRtnuoM
Every 2nd Character: Mutec
○ PS D:\Mountreach_ITR_2026> |
```

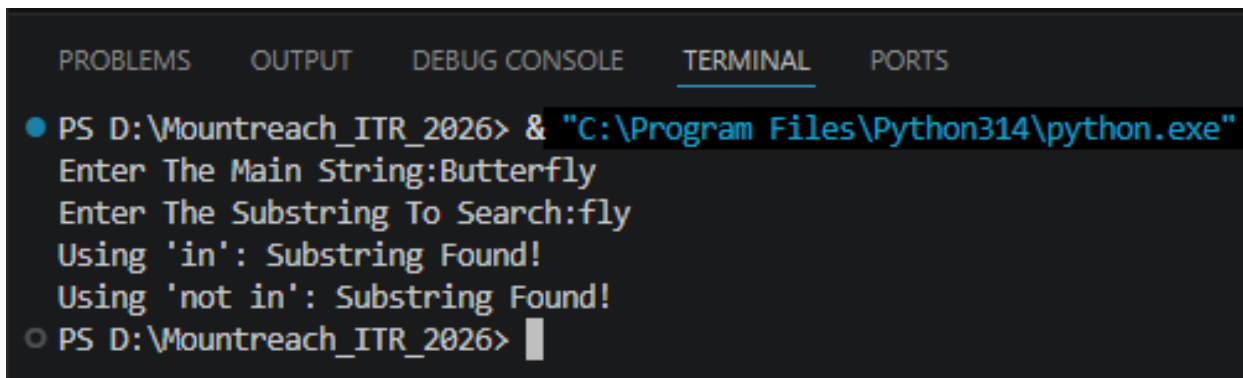
3. String Membership

Write a program that:

- Takes a main string and a substring as input from the user.
- Checks whether the substring is present in the main string using `in` and `not in`.
- Prints appropriate messages (e.g., "Substring found!" or "Substring not found!").

Note that it is case-sensitive.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter The Main String:Butterfly
Enter The Substring To Search:fly
Using 'in': Substring Found!
Using 'not in': Substring Found!
○ PS D:\Mountreach_ITR_2026> |
```

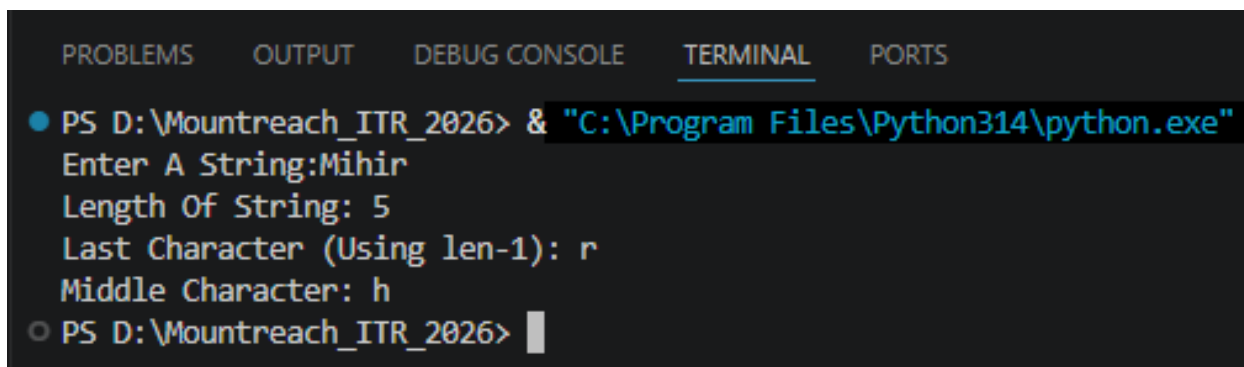
4. len() Function

Write a program that:

- Takes a string as input.
- Prints the length of the string using len().
- Prints the last character using len() (without using negative index).
- Prints the middle character if the length is odd.

Add comments on common mistakes with len().

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:Mihir
Length Of String: 5
Last Character (Using len-1): r
Middle Character: h
○ PS D:\Mountreach_ITR_2026> 
```

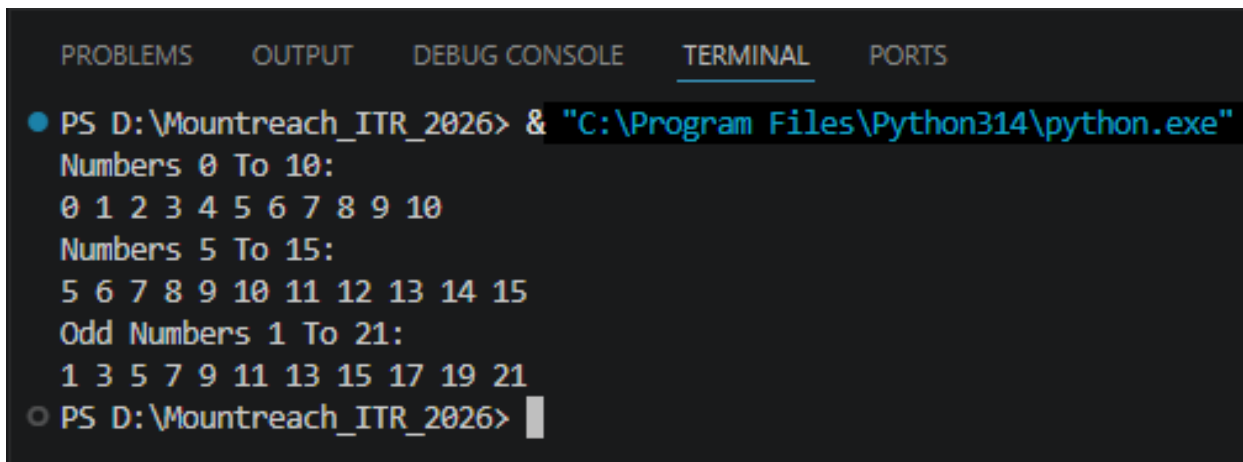
5. range() - Basic Forms

Write a program using for loop and range() to:

- Print numbers from 0 to 10.
- Print numbers from 5 to 15.
- Print odd numbers from 1 to 21.

Explain the three forms of range() in comments.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Numbers 0 To 10:
0 1 2 3 4 5 6 7 8 9 10
Numbers 5 To 15:
5 6 7 8 9 10 11 12 13 14 15
Odd Numbers 1 To 21:
1 3 5 7 9 11 13 15 17 19 21
○ PS D:\Mountreach_ITR_2026> 
```

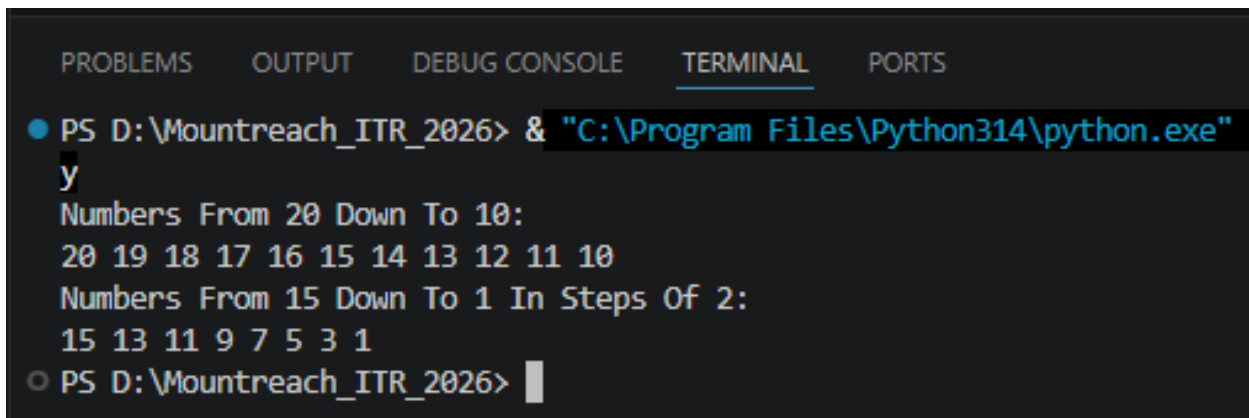
6. range() With Negative Step

Use range() to print:

- Numbers from 20 down to 10 (decreasing).
- Numbers from 15 down to 1 in steps of 2.

Print each sequence on a new line with proper messages.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
y
Numbers From 20 Down To 10:
20 19 18 17 16 15 14 13 12 11 10
Numbers From 15 Down To 1 In Steps Of 2:
15 13 11 9 7 5 3 1
○ PS D:\Mountreach_ITR_2026> 
```

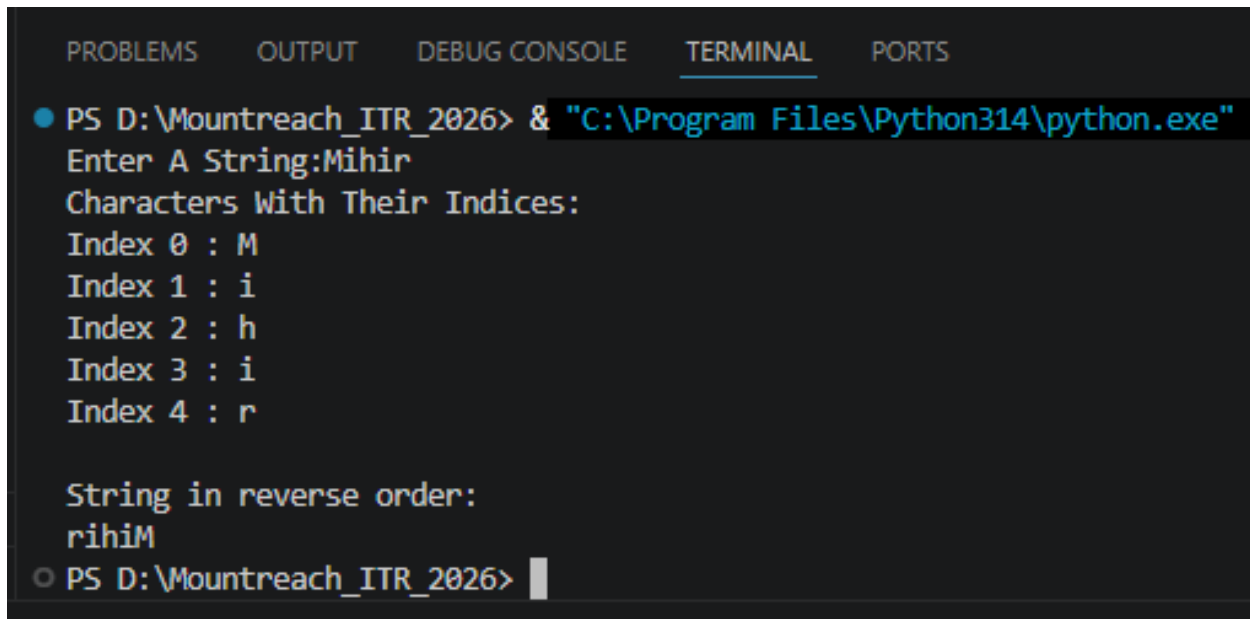
7. Combined: Strings + range()

Write a program that takes a string from the user.

Using a for loop and range() with len(), print:

- Each character of the string one by one (with its index).
- The string in reverse order using negative step in range().

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:Mihir
Characters With Their Indices:
Index 0 : M
Index 1 : i
Index 2 : h
Index 3 : i
Index 4 : r

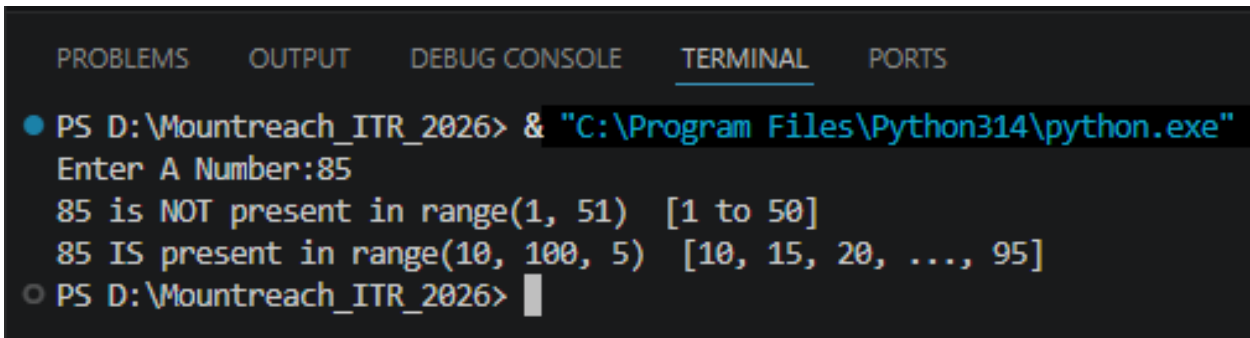
String in reverse order:
rihiM
○ PS D:\Mountreach_ITR_2026> |
```

8. Membership With range()

Write a program that:

- Takes a number as input from the user.
- Checks if the number is present in range(1, 51) and in range(10, 100, 5).
- Prints clear messages for each check.

Output:



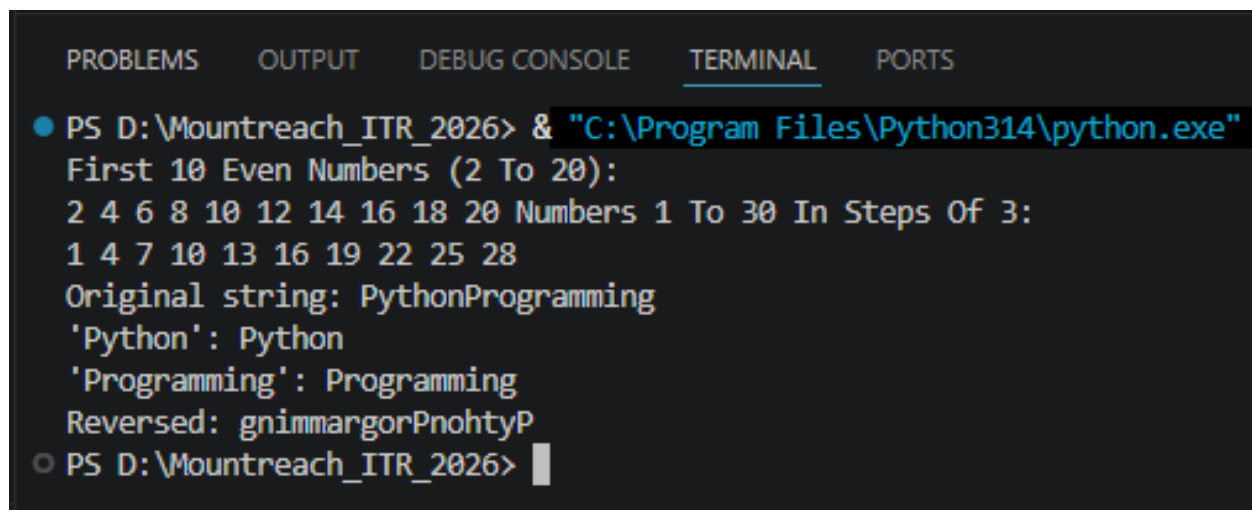
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A Number:85
85 is NOT present in range(1, 51) [1 to 50]
85 IS present in range(10, 100, 5) [10, 15, 20, ..., 95]
○ PS D:\Mountreach_ITR_2026> █
```

9. Slicing + range()

Write a program that prints the following using range() and string slicing concepts where applicable:

- First 10 even numbers (2 to 20).
- Numbers from 1 to 30 in steps of 3.
- A string "PythonProgramming" sliced to show "Python", "Programming", and the reverse.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
First 10 Even Numbers (2 To 20):
2 4 6 8 10 12 14 16 18 20 Numbers 1 To 30 In Steps Of 3:
1 4 7 10 13 16 19 22 25 28
Original string: PythonProgramming
'Python': Python
'Programming': Programming
Reversed: gnimmargorPnohtyP
○ PS D:\Mountreach_ITR_2026> 
```

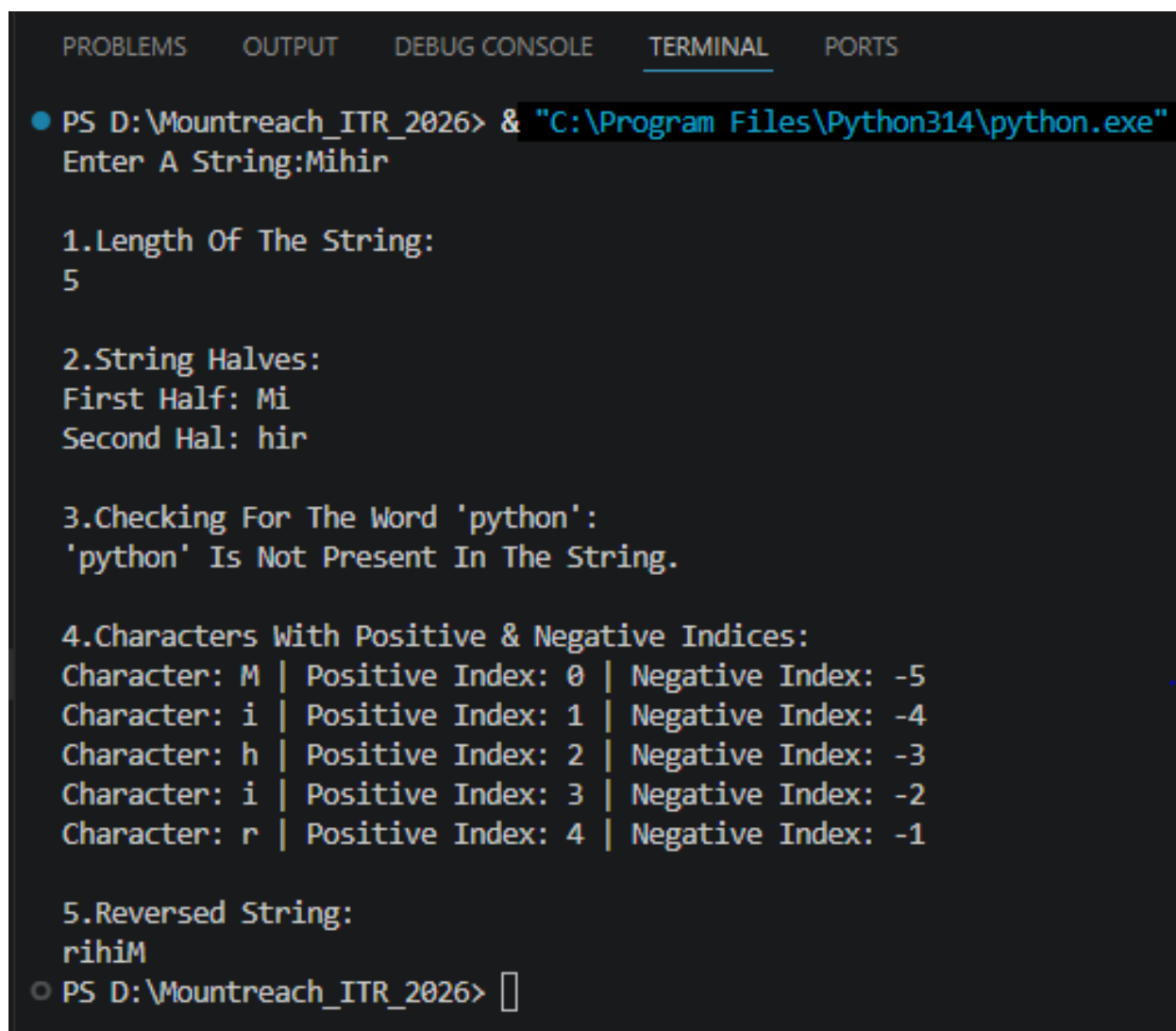
10. Mini Project - Combined

Create a **String Analyzer** program:

- Take a string input from the user.
- Using functions or directly:
 - a) Print length of the string.
 - b) Print first half and second half using slicing.
 - c) Check if the word "python" (case insensitive — you may convert) is present.
 - d) Use range() to print all characters with their positive and negative indices.
 - e) Print the string in reverse.

Add proper comments and use meaningful variable names.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:Mihir

1.Length Of The String:
5

2.String Halves:
First Half: Mi
Second Hal: hir

3.Checking For The Word 'python':
'python' Is Not Present In The String.

4.Characters With Positive & Negative Indices:
Character: M | Positive Index: 0 | Negative Index: -5
Character: i | Positive Index: 1 | Negative Index: -4
Character: h | Positive Index: 2 | Negative Index: -3
Character: i | Positive Index: 3 | Negative Index: -2
Character: r | Positive Index: 4 | Negative Index: -1

5.Reversed String:
rihiM
○ PS D:\Mountreach_ITR_2026> □
```